Safety in Robotic Haircutting

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SURVEY ARTICLE

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Abstract

Robotic haircutting represents an emerging application of service robotics where safety is paramount. Unlike conventional collaborative or domestic tasks, haircutting requires prolonged operation in close proximity to the human head, one of the most vulnerable body regions, while relying on sharp or heated tools. These characteristics make safety concerns central not only to functional reliability but also to user acceptance. This paper reviews the safety landscape of haircutting robots by examining risks across the entire workflow and synthesizing lessons from related domains such as collaborative robotics, healthcare robotics, and autonomous systems. A three-layer framework—mechanical, sensor, and algorithmic—is used to organize existing approaches and identify how safety can be embedded from hardware design to control logic. Highlights opportunities and unresolved challenges in ensuring safe human–robot interaction in haircutting scenarios, providing guidance for future research and deployment.

KEYWORDS
Human-robot interaction, haircutting robots, service robots, human safety

1 | INTRODUCTION

Human–robot interaction (HRI) has become a central research topic as robots increasingly operate in collaborative, healthcare, and domestic environments. Ensuring safety in such interactions is critical, since physical contact may involve sensitive body regions where even minor design flaws can lead to discomfort or injury. Consequently, safety considerations have been emphasized in both engineering practice and regulatory development.

Existing studies on safe HRI have primarily focused on collaborative industrial robots, healthcare robots, and service robots. On the one hand, collaborative robots highlight intrinsic mechanical compliance, force limiting, and standardized safety modes to enable shared workspaces with human operators [1]. On the other hand, healthcare and rehabilitation robots introduce compliant actuation, adaptive control, and fault detection to safeguard patients in close-contact scenarios [2]. Moreover, service robots address ergonomic design, human-aware navigation, and multimodal perception to achieve dependable operation in unstructured environments [3]. These directions form a valuable foundation but remain general in scope.

Haircutting robots represent a distinctive class of service robots where safety requirements are uniquely demanding [4]. Unlike most collaborative or domestic applications, haircutting tasks require continuous and precise manipulation around the head, which is among the most sensitive body regions [5]. In addition, haircutting involves diverse sharp or heated tools, including clippers, scissors, and dryers, that pose direct physical hazards [6]. Beyond physical safety, haircut outcomes must also meet aesthetic requirements, introducing task-specific constraints that are rarely present in conventional HRI domains. These characteristics make robotic haircutting a particularly challenging and underexplored application.

Despite progress in HRI research, systematic approaches to haircutting safety are still missing. There is no unified framework to capture the multiple layers of risks inherent in this task, making it difficult to comprehensively assess and mitigate hazards. To address this gap, this paper provides the first systematic review of safety in robotic haircutting, highlighting the unique challenges of operating around sensitive head regions with sharp or heated tools. A structured analysis of risks across the haircutting workflow is conducted, consolidating hazards from perception, planning, execution, and feedback stages. Building on these insights, a three-layer perspective—mechanical, sensing, and algorithmic—is proposed to organize safety strategies and illustrate how solutions from related domains can be adapted to haircutting robots. This review further identifies open research gaps and points to directions that can guide future studies, experimental validation, and the development of domain-specific standards toward safe real-world deployment.

The remainder of this paper is organized as follows. Section 2 reviews background and related work on HRI safety, highlighting prior studies in perception, hardware compliance, and human factors. Section 3 analyzes the workflow of haircutting robots and categorizes safety risks across stages and system components. Sections 4 to 6 provide an in-depth discussion of the proposed three-layer safety framework: the mechanical layer, the sensing layer, and the algorithmic layer. Section 7 identifies current research gaps and outlines directions for future work. Finally, Section 8 concludes the paper.

2 | BACKGROUND AND RELATED WORK

Safety in HRI has been investigated through multiple complementary perspectives, as robots are increasingly expected to operate in anthropic and unstructured environments. At an early stage, an early atlas of HRI was provided by[7], emphasizing mechanical design, actuation, and control architectures as critical determinants of safety and dependability, and highlighting the need for novel evaluation metrics beyond cognitive-level considerations. In parallel,[8] surveyed the state of the art on safe and dependable HRI, stressing an integrated view that combines mechanical compliance, intrinsic safety design, fault tolerance, and sensing reliability within the PHRIDOM and PHRIENDS projects. More recently, the study in[9] examined the role of autonomy in HRI, differentiating shared control from shared autonomy paradigms and highlighting the risks involved in arbitrating human and robot authority, especially in dynamic or uncertain environments. In addition, some studies have analyzed injury mechanisms and severity indices, borrowing criteria from automotive crash testing such as the Head Injury Criterion (HIC), and extending them to robotic contacts with different body parts[10,11].

Across these studies, common concerns can be identified: mechanical properties often determine the severity of physical contact, control and autonomy strategies are essential for regulating interaction forces, and system dependability relies on sensing reliability and fault tolerance.

2.1 | Perception and Cognition Safety

In HRI, perception and cognition are primarily directed toward detecting proximity, contact, and human motion to prevent harmful collisions. Early approaches emphasized vision-based human detection[12] and skeletal tracking with RGB-D cameras[13] to estimate body joint positions and ensure minimum safe distances between humans and robots. To address occlusions and sensor noise, recent work has advanced multimodal sensing and fusion methods, integrating vision, depth, tactile, and force/torque sensors to provide redundancy and cross-validation[14,15]. Such redundancy is critical in physical contact scenarios, where failures in detection may result in unsafe interaction forces.

Beyond raw perception, cognition in HRI often requires predicting human trajectories and estimating intent. Human modeling studies, particularly those combining musculoskeletal dynamics and sensorimotor control, have been employed to anticipate human motion and adjust robotic responses in real time[16]. Similarly, multimodal frameworks fusing biosignals (e.g., EMG, EEG) with motion data have been proposed for intent recognition and adaptive assistance. These efforts illustrate a growing trend toward combining low-level sensing with higher-level inference to achieve robust perception–cognition pipelines.

2.2 | Action and Hardware Safety

On the above basis, once perception and cognition establish reliable information about human proximity and intent, the hardware and control layer serves as the final safeguard in HRI. Research has shown that compliance control methods, such as impedance and admittance control, are essential to modulate interaction forces and enable safe execution during collisions[17]. In healthcare and rehabilitation robotics, collision detection and reflexive reactions have been widely investigated to ensure safe operation near sensitive body regions[18].

In parallel to control approaches, mechanical compliance has been introduced through soft robotics, viscoelastic covers, and series elastic actuators, which help absorb impact energy and reduce peak collision forces[19,20,21]. Recent developments in semi-active compliant actuation, such as magneto-rheological fluid-based joints, have demonstrated the ability to combine adaptability with intrinsic safety, offering reliable protection during both static and dynamic collisions[22]. These results confirm that a dual strategy of compliant actuation and adaptive control constitutes an effective paradigm for mitigating risks in HRI.

Nevertheless, even when compliant actuation and adaptive control are technically effective, the practical acceptance of HRI systems further depends on human factors and compliance with systemic regulations.

2.3 | Human and Systemic Factors

Beyond technical measures, the effectiveness of HRI safety also depends on how humans perceive safety and how systems

TABLE 1 Standards and directives relevant to safety in haircutting robots

Standard/Directive	Scope and content	Restrictions/implications for haircutting robots
ISO/TS 15066 [23]	Defines collaborative operation modes (power & force limiting, hand guiding, speed & separation monitoring) and specifies pain threshold limits for different human body regions in collaborative robotics	Cutting tools must respect force/pressure thresholds for sensitive regions (e.g., ear, scalp). Motion planning requires separation distance or compliant actuation near the head.
ISO 13482 [24]	Provides safety requirements for personal care robots, including mobile servant, physical assistant, and person-carrying robots	Haircutting robots fall under "personal care" class; design must ensure no hazardous contact, provide safe stopping functions, and integrate protective measures against entrapment or collision.
ISO 10218 [25,26]	Specifies safety requirements for industrial robots and robot systems, including safeguarding, emergency stop, and collaborative operation principles	Although focused on industrial robots, safety provisions such as emergency stop, safeguarded space, and operator training can be adapted for haircutting robots.
ISO 18646 [27,28]	Establishes performance and safety evaluation requirements for service robots, including navigation, obstacle avoidance, and functional tests	Relevant to haircutting robots as a service robotics category; requires standardized testing of positioning accuracy, safety during human contact, and robustness against unexpected user motion.
IEC 60601 [29]	Establishes general safety principles for medical electrical equipment, including requirements for leakage current, thermal safety, and patient protection	Heating devices (e.g., dryers, curlers) integrated into haircutting robots must comply with electrical and thermal safety limits, preventing burns or electrical hazards.
IEC 80601 [30]	Provides collateral and particular requirements for medical electrical equipment used in home healthcare environments	Haircutting robots intended for domestic use must ensure safe integration into non-professional settings, addressing risks from untrained users and variable home conditions.
ISO 14971 [31]	Defines a systematic process for risk management of medical devices, including hazard identification, risk estimation, evaluation, and control	A structured risk analysis process can be applied to haircutting robots to document hazards, establish mitigation strategies, and justify acceptable residual risk.
EU Machinery Directive 2006/42/EC [32]	Provides mandatory requirements for machinery placed on the EU market, covering design principles, documentation, and conformity assessment	Haircutting robots must undergo CE marking, ensure compliance with essential safety requirements, and address liability allocation in case of injury.

comply with regulatory and organizational frameworks. Human trust and comfort are shaped not only by objective limits on force and pressure but also by subjective factors such as predictability, controllability, and cultural acceptance [33,34,35]. Prior studies have shown that perceived safety may deviate substantially from physical safety, particularly when interaction involves sensitive body regions such as the head and neck [36]. This discrepancy underscores the importance of ergonomic design and transparency of robot behavior in safety evaluations. To capture these interrelated dimensions, Figure 1 illustrates a conceptual framework that integrates regulatory standards, human factors, and systemic issues. In this framework, regulatory standards provide objective references for safety limits, human factors determine the subjective acceptance of the public to the robots, and systemic issues address deployment aspects such as certification, liability, and maintenance. The interaction of these three domains ultimately determines whether physical safety can be effectively translated into user trust and social acceptance.

Building upon this conceptual view, Table 1 summarizes the main international standards and directives that shape safety requirements for haircutting robots. Standards such as ISO/TS

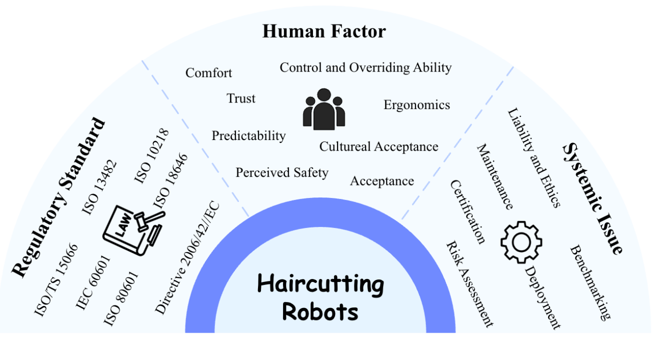

FIGURE 1 Conceptual framework of human and systemic factors for haircutting robots. Safety depends on the integration of regulatory standards, human factors, and systemic issues.

15066, ISO 13482, and ISO 10218 address collaborative operation, personal care robotics, and industrial robot safety, while ISO 18646 and IEC 80601 extend requirements to service and domestic environments. In addition, ISO 14971 and the EU Machinery Directive provide structured approaches to risk management, certification, and liability allocation. Together, these standards define both the technical criteria for robot safety

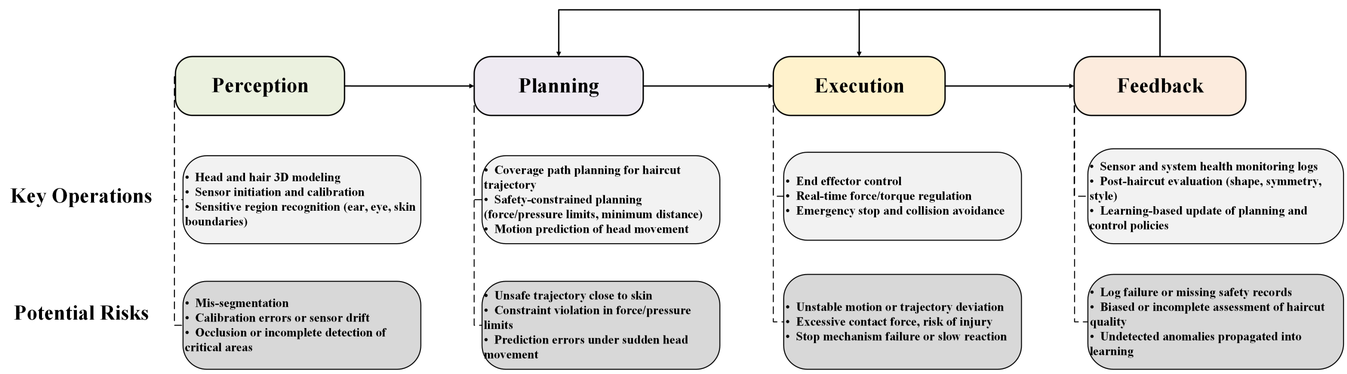

FIGURE 2 Workflow stages of haircutting robots with key operations and potential risks. Each stage—perception, planning, execution, and feedback—includes representative tasks and associated safety risks, illustrating the progression from sensing to decision-making, action, and post-operation learning.

and the procedural obligations for compliance in real-world deployment.

In summary, perception and cognition, action and hardware, and human as well as systemic considerations represent three interconnected perspectives of HRI safety. Prior studies in collaborative, service, and medical robots provide foundational principles, including multimodal sensing, compliant control, soft materials, and standardized force thresholds. However, haircutting robots simultaneously face unique challenges across all three domains: fine-grained perception of hair and skin, continuous close-contact manipulation with diverse tools, and heightened user sensitivity in head interactions. These distinctive conditions underline the necessity of a tailored safety framework for haircutting robots.

3 | RISK LANDSCAPE AND SYSTEM WORKFLOW

3.1 | Workflow Analysis and Risk Categorization

The haircutting process of robots can be structured into four successive stages: perception, planning, execution, and feedback. Each stage involves critical operations but also introduces potential safety risks, as illustrated in Figure 2. Understanding this workflow is essential to identify where failures may occur and how they propagate across the system.

In the **perception** stage, the robot acquires geometric and semantic information of the head and hair through multimodal sensors. Errors in segmentation, calibration drift, or occlusions may lead to inaccurate recognition of sensitive regions such as the ears or scalp. If not corrected, these perception errors directly compromise subsequent planning.

During the **planning** stage, hairstyle models are combined with safety constraints to generate feasible cutting trajectories. Risks arise when trajectories are computed too close to the skin, when human motion is not adequately predicted, or when algorithmic errors bypass safety margins. Additional vulnerabilities occur in teleoperated settings, where network latency or cyberattacks may modify or delay trajectory commands.

In the **execution** stage, the robotic manipulator operates clippers, scissors, or heating tools in direct contact with the user. Physical hazards include excessive contact forces, unsafe blade geometry, overheating of dryers or curlers, and unintended collisions between the robotic arm and the head. These risks are critical because they involve immediate physical interaction with the user.

Finally, the **feedback** stage integrates sensor monitoring and post-operation assessment. Failures in this stage, such as delayed anomaly detection or incomplete task confirmation, may allow hazardous actions to continue without correction. Inadequate feedback also prevents proper adaptation and learning for subsequent tasks.

Overall, risks can be abstracted into five overarching categories: network and cybersecurity, perception and sensing, physical and tool-related, control and algorithmic, and human-related. In addition, Table 2 consolidates these categories by aligning typical issues in haircutting scenarios with the associated system components. This categorization reveals that risks are not confined to a single function but rather distributed across communication infrastructure, sensing hardware, physical tools, control algorithms, and human interaction.

The presence of physical risks stemming from tools and actuators emphasizes the role of inherent safety design. Perception-related vulnerabilities highlight the importance of redundant sensing and monitoring. Control instabilities and cyberattacks demonstrate the necessity of robust decision-making and supervisory mechanisms.

TABLE 2 Categorization of risks in haircutting robots across system aspects

Risk category	Typical issues	Associated components
Network & cybersecurity	• High latency in teleoperation causing delayed reaction • Network congestion leading to packet loss • Cyberattacks that hijack robot control • Unauthorized modification of haircut trajectory • Service interruption from unstable wireless connection	Communication link, wireless modules, teleoperation interface, network protocols
Perception & sensing	• Misclassification of hair, skin, or ear boundaries • Depth camera errors due to lighting or occlusion • Force/torque sensor drift or calibration failure • Loss of vibration or acoustic signals during tool–hair interaction • Single-sensor dependency without redundancy	Cameras, depth sensors, force/torque sensors, acoustic/vibration sensors, sensor fusion modules
Physical & tool-related	• Excessive contact force from clippers or scissors • Unsafe blade geometry causing skin injury • Overheating of dryers or curlers near scalp • Accidental collision with robot arm or end-effector • Tool wear leading to irregular performance	Clippers, scissors, heating tools (dryer/curler), blade geometry, robotic arm end-effector
Control & algorithmic	• Instability of controller under time delay • Unsafe motion planning without safety margins • Lack of fallback strategies when sensors fail • Runtime monitoring failure to stop hazardous action • Inconsistent tool switching logic	Motion planner, obstacle avoidance module, controller, runtime safety monitor, tool-switching logic
Human-related	• Sudden head movements (rotation, tilting, reflexes) • Postural shifts due to discomfort or fatigue • Fear or psychological stress during close contact • Non-cooperation or unexpected user actions • Lack of trust or acceptance of robotic haircutting	User head, posture, facial regions (ear/neck), behavioral response, psychological state

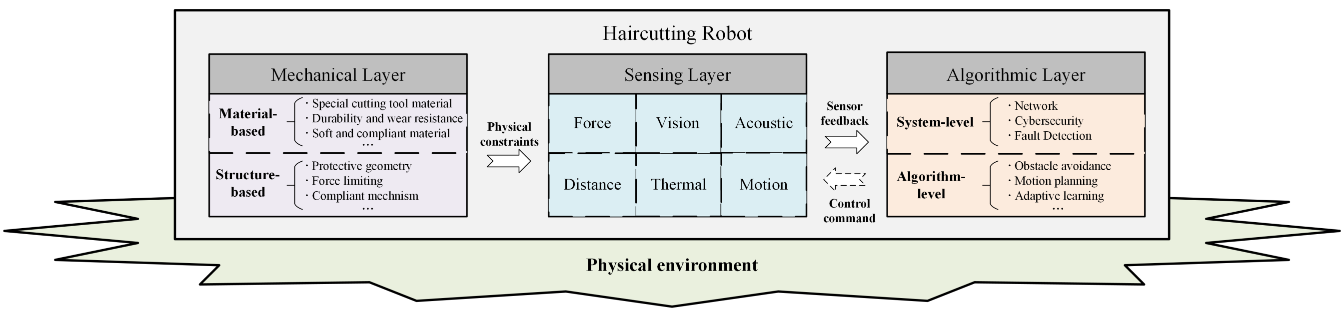

FIGURE 3 Three-layer safety framework for haircutting robots with layer-wise components

Based on the above analysis, several key questions arise during the haircutting process:

1. How can the robot accurately detect the detailed geometry of the human head and the dynamic environment?
2. Once this information is obtained, how can the robot execute the required actions safely?
3. If an unexpected event occurs, how can injury to the user be reduced or avoided?

To address these challenges, a three-layer safety framework is proposed to systematically mitigate risks in haircutting robotic systems, as illustrated in Figure 3. The **mechanical layer** provides baseline protection through compliant design, guarded tool geometry, and safe material selection, ensuring that unintended physical contact does not result in severe harm. The **sensing layer** integrates multimodal sensing and redundancy, enabling reliable detection of head geometry, user motion, and abnormal interactions. The **algorithmic layer** governs motion planning, controller stability, and cybersecurity defenses, ensuring that task execution remains safe even under uncertainty or system delays.

These layers are interconnected rather than independent. The mechanical layer establishes the physical boundaries within which perception operates reliably. The perceptual layer supplies real-time information that constrains and guides algorithmic decisions. In turn, the algorithmic layer supervises both mechanical actuation and sensing feedback, enforcing

TABLE 3 Real-world mechanical incidents and issue analysis

Instance	Description	Issue analysis
Instrument tip-cover over-advancement in robotic surgery [37]	Tip-cover accessory advanced too far, increasing scissor diameter and preventing smooth insertion/removal from cannula; no error alert	Assembly misalignment and poor tolerance design
Robotic surgical arm collisions triggering "power limit exceeded" [38]	da Vinci system arms collided in certain cases, exceeding power output limits due to rigid structure and workspace overlap	Excessive arm rigidity and insufficient collision buffering
Review of da Vinci surgical system failures [39]	Among 1797 operations, 43 (2.4%) had malfunctions, including arm, optical, and instrument failures, often linked to wear, fatigue, or poor durability	Material fatigue and inadequate structural durability
Force/pressure thresholds in human–robot collisions [40]	Experiments on pigskin and other tissues measured pain thresholds under impacts; highlighted role of shape, surface hardness, and edge design in injury severity	Hard surfaces and sharp edges lacking compliant materials

safety policies and corrective actions. Through this hierarchical and interdependent structure, safety responsibilities are distributed across hardware, sensing, and control levels, reducing the risk of systemic failure and avoiding reliance on any single safeguard. Sections 4–6 provide detailed discussion of each layer.

4 | MECHANICAL LAYER

The mechanical layer forms the fundamental basis of the safety framework in robitc haircutting, as it defines the physical materials and structural geometry of the system. It provides the first line of defense for user safety by shaping external interfaces, embedding passive protections, and incorporating inherent compliance. Mechanical safety is particularly critical because any malfunction or misdesign at this layer cannot be fully compensated by higher-level sensing or control.

4.1 | Mapping from Real-world Incidents to Haircutting Scenarios

Analysis of real-world incidents summarized in Table 3 shows that failures at the mechanical layer often result in direct and severe consequences. Assembly misalignments, such as mispositioned accessories, demonstrate how even small tolerance errors can immediately compromise safety without leaving room for software correction. Arm collisions caused by excessive rigidity highlight that control responses are insufficient once unsafe forces are already transmitted. Long-term fatigue and wear underline that material degradation silently reduces reliability until sudden breakage occurs. Moreover, studies on force thresholds confirm that surface geometry and hardness are decisive factors for injury severity, regardless of force limits imposed by algorithms.

These findings emphasize the critical role of mechanical protection: once hardware design or materials are inadequate, sensing and control layers cannot fully prevent harm. For haircutting robots, the same patterns can be projected. Misassembly of clipper guards could expose blades, leading to accidental cuts. Rigid manipulators without compliance may cause bruises if the customer moves unexpectedly. Fatigue of tool heads or joints may result in cracks or detachment close to the scalp, creating sharp fragment hazards. Hard or sharp housing surfaces could injure the skin even under minor contact.

Therefore, lessons from existing robotic systems indicate that the mechanical layer must provide inherent safety margins in both materials and structures. Without such safeguards, haircutting robots risk exposing users to direct physical harm, undermining both functional reliability and user trust.

4.2 | Safety Solutions

To mitigate these risks, safety can be enhanced through two complementary approaches: material-based design and structure-based design (see Figure 4).

A. Material-based safety design

- **Special cutting tool materials:** For example, ceramic blades are favored over conventional metallic blades because of their lower thermal conductivity, higher wear resistance, and reduced cutting aggressiveness [41]. Such material choice decreases the likelihood of accidental skin laceration while simultaneously minimizing heat transfer to the scalp during prolonged operation.
- **Soft and compliant materials:** Surrounding the housing with elastomeric layers (e.g., silicone or TPU, Shore A 20–40) allows localized impact energy to be absorbed and contact pressure to be distributed. According to ISO/TS 15066, the allowable transient contact force for the human head

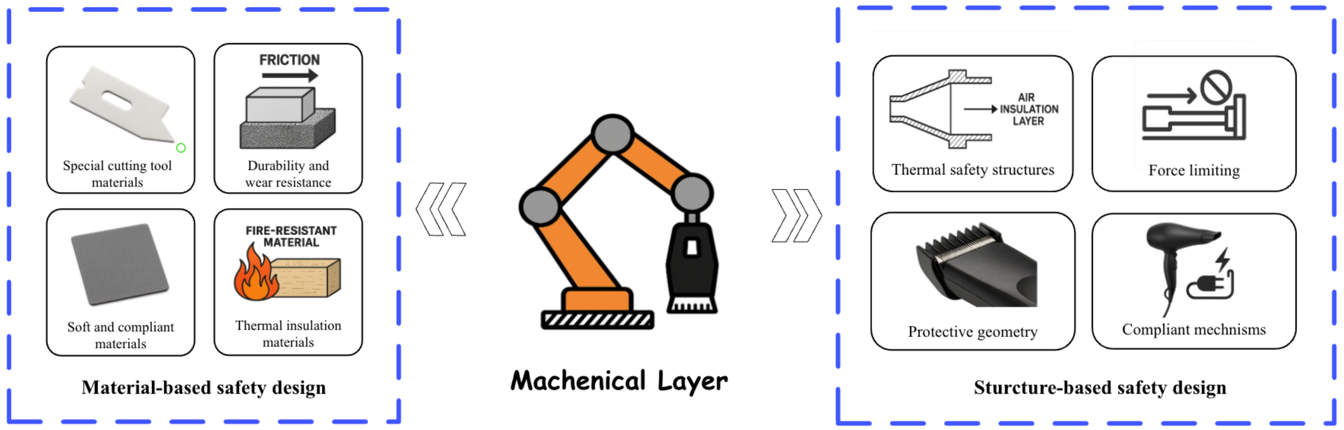

FIGURE 4 Mechanical-layer safety strategies categorized into material-based and structure-based designs

is approximately 65 N, with pressure thresholds around 90 N/cm^2 [42]; compliant layers help maintain contact within these safe ranges.

- **Thermal insulation materials:** The use of heat-resistant polymers and ceramic barriers, combined with double-wall housings containing air gaps, ensures thermal isolation of blow dryers or curling devices. These passive structures limit surface temperature rise below 43–45 °C [43], corresponding to safe exposure thresholds for human skin.
- **Durability and wear resistance:** Surface coatings such as diamond-like carbon (DLC) or polytetrafluoroethylene (PTFE) reduce blade friction, improve resistance to corrosion from hair care products, and extend service lifetime. Some studies in the hair clipper/blade domain have shown significantly improved durability under cyclic usage, though cutting-cycle counts vary depending on usage and testing methods. For example, DLC coatings have reduced volumetric wear by up to 75-99 % in simulated body/physiological environments [44].

B. Structure-based safety design

- **Protective geometry:** Protective geometry can be considered from two complementary perspectives. First, tool-level design focuses on the geometry of the cutting element itself, where rounded blade edges and increased edge radius reduce penetration risk compared to sharp tips below 10 μm [50], and blunt edge shaping further decreases the likelihood of laceration [51]. Second, structural-level design introduces additional barriers such as distance combs, foil guards, or dual-layer structures (e.g., comb teeth plus transparent screens), which allow hair to enter the cutting zone while preventing skin intrusion [52]. These complementary approaches create redundant safety boundaries that passively protect the scalp even in cases of sudden user movement.
- **Force limiting mechanisms:** Mechanical devices such as slip clutches, torque fuses, or magnetic couplers are designed to disengage the drive once transmitted torque exceeds a threshold [53,54]. This prevents excessive contact forces on the scalp in scenarios of tool jamming or abrupt external disturbance, offering an inherent safety margin independent of control response time.
- **Compliant mechanisms:** Series elastic couplings integrated at structural joints provide inherent compliance by absorbing impact forces [55]. Even if the robot contacts the user unexpectedly, the resulting interaction forces remain within tolerable ranges before higher-level intervention occurs [56].
- **Thermal safety structures:** Double-wall nozzles with integrated air insulation and passive thermal cut-off devices, such as bimetallic switches, act as fail-safe protections in heated styling modules. For example, early hair dryer designs included nozzles with inner and outer walls and an overload thermal switch placed within the nozzle structure, which interrupts heating when excessive temperatures are reached (US Patent 3,943,329 [57]). Commercial bimetallic thermal switches (e.g. ST-12) are used in present-day hair dryers for passive temperature limiting [58]. These mechanisms guarantee that the temperature at the scalp interface does not exceed safety thresholds even in the event of control malfunction or prolonged stall conditions.

This design from mechanical-layer safety directly shape the feasibility of future haircutting robots. Effective choices in materials and structural configurations not only reduce the likelihood of injuries but also enhance comfort, reliability, and

TABLE 4 Real-world sensor-layer incidents and issue analysis

Instance	Description	Issue analysis
Industrial robot in Korea misidentifies worker as a box [45]	Robot compressed an inspector after classifying him as cargo	Insufficient sensing modalities and lack of sensor-fusion safeguards
Boeing 737 MAX crashes due to faulty AOA sensor [46]	Angle-of-attack sensor error triggered inappropriate nose-down commands; two crashes	Single-sensor dependency without redundancy or cross-validation algorithms
Waymo car with occluded sensors halts abnormally [47]	Post-minor-incident tape covered LiDAR/cameras; environment not perceived correctly	No fallback sensing or occlusion-detection mechanisms
Uber AV fails to recognize pedestrian, fatality [48]	System failed to timely detect a pedestrian walking a bicycle, no effective braking	Limited low-light sensing capability and inadequate sensor calibration
da Vinci endoscope fogging / focus loss reported in MAUDE [49]	Endoscopic lens fogging or focus issues reduced visibility, contributing to errors	Optical sensor degradation under environment without real-time compensation

user acceptance in close-contact tasks around the head. By embedding inherent safety into the hardware itself, designers can lower the dependency on complex perception and control safeguards, creating systems that are more robust to uncertainties in human motion and environmental variability. This establishes a practical pathway toward deploying haircutting robots in everyday service contexts with both functional reliability and public trust.

5 | SENSING LAYER

Building on the inherent protections of the mechanical layer, the sensing layer extends safety into the domain of perception. While structural design constrains physical risks, sensors determine whether the robot can adapt to human variability and environmental uncertainty in real time. In haircutting tasks, where close contact with the head is inevitable, the ability to reliably sense force, motion, distance, and temperature forms the foundation for proactive risk management. This chapter focuses on how sensing systems contribute to safety by transforming raw measurements into reliable indicators of interaction state.

5.1 | Mapping from Real-world Incidents to Haircutting Scenarios

Analysis of the incidents summarized in Table 4 indicates that sensor shortcomings translate rapidly into unsafe behavior. First, insufficient sensing modalities and absent fusion safeguards allow critical misidentifications, such as a human classified as cargo, showing that single streams are brittle to appearance variation. Second, single-sensor dependency without redundancy or cross-validation (e.g., faulty AOA input) can drive the system into hazardous states even when all downstream logic operates nominally. Third, unhandled occlusion and lack of fallback sensing demonstrate that perception can collapse to unusable states after minor disturbances. Fourth, limited low-light capability and inadequate calibration degrade detection timeliness and confidence precisely when fast reaction is needed. Finally, optical degradation without real-time compensation (like fogging, soiling, and defocus) reduces effective visibility and silently erodes safety margins.

When mapped to haircutting robots, the implications are direct. Relying on a single RGB camera risks ear/skin misclassification under hairstyle and complexion diversity; a fusion design with depth, proximity, or force cues is required to bound tool–skin clearance. Any single-point sensor failure (e.g., a sole proximity sensor drifting) could remove contact guards and permit unsafe approach; dual-channel sensing with cross-checks should be mandated for clearance and force. Occlusion events—hair tufts, hands, or accessories blocking the lens—must be detected and handled with fallback proximity/force supervision to avoid blind operation. Low-light or miscalibrated cameras can delay boundary detection near the temple and nape; on-device illumination, periodic self-calibration, and confidence-aware gating are needed to prevent late reactions. Environmental optical degradation (spray mist, steam, sebum) should trigger cleaning prompts or automatic derating, while thermal and vibration side channels monitor for emerging hazards. Therefore, these mappings show that sensor-layer safety in haircutting robots hinges on modality diversity, online health monitoring, and explicit fail-safe behaviors whenever sensing confidence drops.

5.2 | Safety Solutions

Sensing devices form the backbone of safety assurance in haircutting robots, as they provide real-time information about contact forces, motion dynamics, spatial boundaries, temperature, and tool conditions. These signals enable the robot to enforce limits on interaction, detect abnormal events, and maintain safe operating conditions.

A. Safety assurance by individual sensors

At the sensing layer, devices can be categorized by the type of data they provide, as each modality contributes specific mechanisms for hazard prevention:

Force and current sensing underpin safe physical interaction. Motor currenxt signals allow indirect estimation of joint torques[59], while six-axis force/torque sensors, strain-gauge load cells, piezoresistive pads, and capacitive or optical skins provide direct contact measurements[60]. In haircutting robots, these measurements ensure that pressing forces remain within tolerable limits. For instance, when a clipper guard becomes misaligned and exposes the blade, abnormal torque or local contact pressure can be detected and used to trigger immediate motor shutdown. Distributed tactile pads further localize contact points around sensitive regions such as the ear or temple, where excessive force could otherwise result in bruising or laceration[61].

Motion and acceleration sensing enable rapid collision detection. Inertial measurement units (IMUs), consisting of accelerometers and gyroscopes, capture sudden kinematic changes: high-frequency accelerometers detect sharp impacts, while gyroscopes identify unexpected angular deviations[62]. In haircutting robots, these signals are essential for handling unpredictable head motion by the customer. For example, if the head rotates suddenly while the robot is trimming near the ear, accelerometer spikes can trigger an emergency stop, while gyroscope readings allow trajectory adjustment to prevent scraping. The short response time minimizes exposure to hazardous contact[63].

Vision and distance sensing provide spatial awareness. RGB and stereo cameras, structured-light, and time-of-flight (ToF) systems detect boundaries between hair, skin, and ears[64,65,66]. For near-field safety, infrared, ultrasonic, or optical reflectance sensors enforce minimum tool–scalp distances. In haircutting robots, this dual-layer perception ensures that even if hair partially occludes the visual field, proximity sensors can still guarantee that cutting blades do not directly touch the scalp[67]. Furthermore, vision-based algorithms can identify the contour of ears and prevent unintended clipping injuries, a common source of discomfort in manual haircuts.

Thermal sensing safeguards heated tools. Thermistors, thermocouples, resistance temperature detectors (RTDs), and infrared thermopiles measure point temperatures, while thermal cameras or multi-pixel arrays capture broader temperature distributions[68,69,70]. In haircutting robots that integrate blow dryers or curling devices, these sensors enforce automatic power reduction or cut-off when surface temperatures approach human tolerance levels. This prevents scalp burns in scenarios where airflow is blocked by hair or the tool is held too close to the skin for prolonged periods. Thermal imaging can further detect localized hot spots, providing redundancy against failures of single-point sensors.

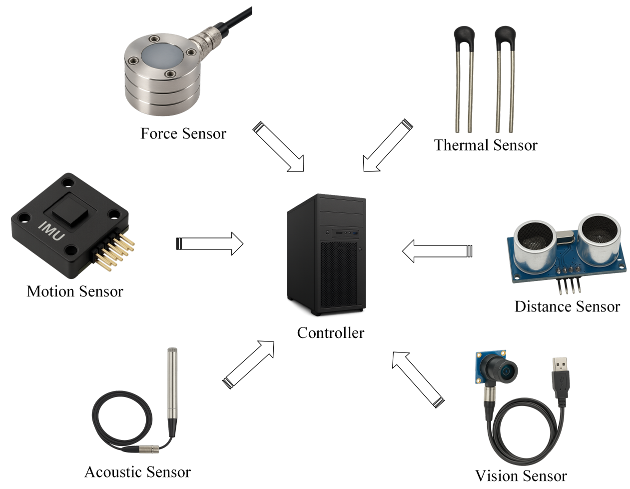

FIGURE 5 Multimodal sensor fusion framework for haircutting robots, enabling complementary information to be combined for robust perception.

Vibration and acoustic sensing detect anomalies that precede mechanical hazards. MEMS accelerometers, piezoelectric sensors, and microphone arrays identify abnormal oscillations, grinding sounds, or resonance shifts caused by blade wear, imbalance, or entanglement with hair[71,72]. In haircutting robots, such early warnings are critical to prevent escalation into dangerous events. For instance, if hair entangles within the blade assembly, the acoustic signature changes sharply; detecting this pattern enables the system to disengage the motor before the jam translates into pulling, skin pinching, or fragment release[73].

However, each modality has inherent limitations. Vision can be affected by hair occlusion or illumination; force sensors add cost and weight; IMUs are noise sensitive; thermal sensors may respond too slowly; vibration and acoustic sensing are prone to background interference. In the complex environment of haircutting—characterized by fine hair structures, reflective surfaces, humidity, and unpredictable head motion—relying on a single sensor type is insufficient.

B. Redundancy and multimodal integration

Although single sensors provide essential functions, haircutting robots must operate under variable lighting, hair occlusion, humidity, and unpredictable motion. Robust perception therefore relies on redundancy and multimodal fusion.

Sensor redundancy ensures fault tolerance by monitoring critical variables through overlapping modalities. For example, scalp clearance can be verified simultaneously by

TABLE 5 Real-world algorithm/control-layer incidents and issue analysis

Instance	Description	Issue analysis
Uber ATG Tempe fatal crash [74]	NTSB found the ADS failed to correctly classify/predict the pedestrian and emergency braking was disabled in autonomy, relying on a distracted safety driver	Incomplete perception–prediction algorithm and disabled safety override logic
Tesla Autopilot crashes into stationary vehicles (EA22-002) [75]	NHTSA aggregated >200 frontal-plane crashes, including first-responder scenes with stationary obstacles, under driver-assist engagement	Algorithmic bias toward moving objects; inadequate collision-avoidance software
FDA Class II recall: da Vinci Xi software anomaly [76]	FDA notice reported that a software anomaly could cause unexpected master movement and unintended instrument tip motion	Software defect in control logic without fail-safe routines
Teleoperated surgical robot cyberattacks (Raven II) [77]	Researchers demonstrated that packet manipulation/DoS attacks could induce unsafe motions or cause command loss in a teleoperated surgery robot	Lack of secure communication protocols and software-level integrity checks
Telesurgery delay limits and performance degradation [78]	Reports indicated that large delays ($\geq$500–750 ms) sharply increase errors and can render direct teleoperation unsafe	Control algorithms not robust to high-latency networks
Tesla Autopilot misclassifies white trailer as sky [79]	Autopilot failed to detect a white semi-trailer crossing the road; no braking, fatal crash	Faulty perception algorithm and misconfigured sensor-fusion filters

proximity sensors and cameras, while both current-based torque estimation and direct force sensing offer independent measures of contact load [80]. This redundancy ensures that safety-critical variables remain observable even if one sensor is obstructed by hair, degraded by cosmetic residues, or temporarily malfunctioning [81].

Multimodal fusion enhances robustness by combining heterogeneous signals (see Figure 5). Fusing IMU data with acoustic features enables rapid detection of tool jamming or entanglement, while integrating vision with force sensing distinguishes intentional hair trimming from accidental skin contact [82]. Such fusion compensates for environmental variability, reduces the need for intrusive sensor placement, and aligns perception outputs with established human safety thresholds [83]. By leveraging complementary information, multimodal perception provides a more reliable basis for initiating protective actions.

Overall, redundancy and multimodal fusion transform isolated sensing modalities into a coherent perceptual shield. By continuously constraining forces, monitoring motion, enforcing clearances, regulating temperature, and detecting anomalies, the sensing layer enables haircutting robots to identify and mitigate risks in real time.

The sensing layer thus bridges the gap between physical safeguards and algorithmic control. By consolidating heterogeneous signals into consistent safety indicators, it provides the information basis for higher-level planning and supervisory mechanisms. In this way, perception does not replace mechanical design but complements it, ensuring that safety is actively monitored and communicated to the control layer. The next section builds on this foundation by examining how control strategies enforce safe execution under uncertainty.

6 | ALGORITHMIC LAYER

The mechanical and sensing layers provide the fundamental hardware safety and perceptual capabilities, while the algorithmic layer is ultimately responsible for transforming raw signals and structural constraints into safe and coordinated actions. It governs motion planning, trajectory generation, real-time control, and supervisory logic, thereby serving as the decisive interface between human intent, sensory perception, and physical execution.

6.1 | Real-world incidents and mapping to haircutting robots

Real-world incidents across autonomous driving, surgical robotics, and teleoperation reveal recurring vulnerabilities at the algorithmic and control layer (see Table 5). A common theme is that algorithmic errors or missing safeguards, rather than hardware malfunctions, often act as the root cause of safety-critical failures. For instance, crashes involving autonomous vehicles such as the Uber ATG Tempe case and Tesla collisions with stationary objects demonstrate how incomplete perception–prediction pipelines or biased collision-avoidance logic can lead to catastrophic consequences even when mechanical subsystems remain intact. Similar issues have been documented in medical robotics, where recalls of surgical platforms highlighted unexpected actuator motions triggered by software anomalies, underscoring the lack of built-in fail-safe routines. In parallel, studies on telesurgery and teleoperated

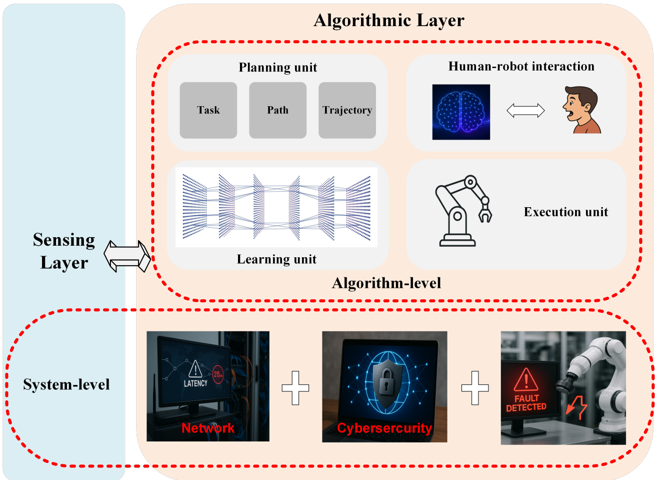

FIGURE 6 Cross-layer safety unit for haircutting robots. The unit integrates network monitoring, cybersecurity, and fault detection as supervisory modules that operate across both sensing and algorithmic processes, ensuring robustness against latency, malicious interference, and hardware failures.

robots have shown that large network delays or deliberate cyberattacks can directly destabilize control, inducing unsafe end-effector motions or preventing timely corrective actions.

When mapped to haircutting robots, these observations indicate that algorithmic fragility could readily translate into hazardous behavior in close-contact human environments. A failure to reliably predict head motion or to maintain skin–tool clearance could expose the scalp or ears to laceration. Software exceptions without supervisory override may cause unintended blade movement near sensitive regions. Latency in remote operation or cloud-based processing could delay responses to sudden user motion, resulting in unsafe contact before correction is possible. Moreover, compromised communication channels could enable malicious manipulation of motion commands, threatening both user safety and trust. These scenarios underline that, in haircutting robots, the algorithmic layer must be designed not only for nominal performance but also for resilience against perception errors, software anomalies, and degraded or adversarial network conditions.

6.2 | Safety Solutions

As shown in Figure 6, safety mechanisms at the algorithmic layer operate at two complementary levels. On the one hand, supervisory modules—such as network reliability, cybersecurity, and runtime fault detection—span perception and control to ensure resilience against timing uncertainties, adversarial interference, and internal faults. On the other hand, safety can be embedded across the algorithmic stack itself, including state estimation and uncertainty quantification, human-motion prediction and intent recognition, decision making and task scheduling, motion planning and control, and data-driven adaptation under explicit safety constraints, so that unsafe actions are precluded throughout the haircut workflow.

A. System-level Reliability and Protection

Network reliability. In teleoperated haircutting scenarios, data transmission latency or congestion can undermine stability and operator control [84]. A two-stage strategy is essential. Before a session begins, the system evaluates round-trip delay and available bandwidth, authorizing operation only when latency is acceptable. During the haircut, continuous monitoring of quality of service ensures that if latency exceeds limits or packet loss rises, the robot either pauses or prompts for local human takeover. From the control perspective, predictive and passivity-based controllers can mitigate delay-induced instability, while control barrier functions (CBFs) enforce safety envelopes around the scalp even under delayed actuation [85,86]. Safe reinforcement learning further constrains policy exploration so that generated trajectories never violate safety limits [87,88].

Cybersecurity. Both teleoperated and autonomous modes are vulnerable to cyberattacks. Encrypted communication, secure authentication, and integrity checks are therefore mandatory [89,90]. In addition, haircut-specific path-integrity safeguards can be employed: planned trajectories are generated and reviewed prior to execution, and during operation the executed path is continuously compared with the reference. If deviations exceed tolerance—whether due to hacking, operator error, or unforeseen anomalies—the robot halts or requests explicit confirmation before proceeding [91]

Fault detection and predictive maintenance. Even with secure communication and stable control, internal faults may propagate into unsafe behavior if undetected. Real-time monitoring of motor currents, vibration, and proximity data allows immediate detection of excessive forces, abnormal oscillations, or sensor degradation, prompting emergency stop [92]. Beyond reactive protection, predictive maintenance methods—such as anomaly detection, statistical health monitoring, and model-based prognostics—anticipate wear, backlash, or drift before they compromise safety, enabling timely intervention [93,94,95].

B. Algorithm-level Control

In addition to ensuring safety via specific unit, it can also be embedded across the algorithmic stack, ensuring that perception, decision-making, and control all operate within protective boundaries:

Obstacle avoidance. Planning algorithms [96,97,98] can incorporate geometric models of the head, ears, and tools, enforcing spatial constraints so that end-effectors never intrude into restricted regions. Online re-planning enables safe trajectory redirection when the user moves unexpectedly.

Predictive motion planning. By forecasting head movement from vision or inertial cues, the robot adjusts tool paths proactively rather than reactively[99,100,101]. Anticipatory planning reduces the need for abrupt emergency stops and provides smoother, safer interaction.

Human–robot interaction. Shared control mechanisms allow the user to influence robot actions through subtle cues, such as head inclination or verbal commands[102]. This integration of user intent promotes comfort and trust while maintaining safety during close-contact operations.

Safe learning and adaptation. Data-driven methods can personalize haircut strategies to different head shapes or hair types, but adaptation must remain safety-constrained. Approaches such as constrained policy optimization or safety filters ensure that exploration and policy updates remain within verified safe bounds[103,104,105,106].

By combining supervisory safeguards with safety-aware algorithmic strategies, the control layer not only mitigates external disturbances and internal faults but also constrains the robot's decision-making and adaptation processes, ensuring predictably safe operation throughout the haircutting workflow.

In conclusion, the algorithmic layer extends safety from passive protection toward proactive assurance. Supervisory modules protect against faults, adversarial influence, and communication uncertainties, while algorithmic strategies embed safety directly into perception, decision making, and control. Together, these mechanisms transform heterogeneous sensor inputs into verifiable safety guarantees, ensuring that the robot adapts predictably to human variability and environmental disturbances. When combined with the mechanical and sensing layers, the algorithmic layer completes a multilayered framework in which safety is enforced not only by physical design and perceptual monitoring, but also by the very logic that governs robot behavior.

7 | RESEARCH GAP AND FUTURE WORK

Beyond the three-layer safety framework proposed in this work, significant research gaps remain in addressing safety challenges unique to haircutting robots. These gaps highlight the need for systematic studies, experimental validation, and eventual standardization before safe real-world deployment can be achieved.

- **Lack of quantified safety thresholds.** Unlike industrial collaborative robots, where safety limits on impact forces and permissible contact pressures have been extensively studied, the haircutting scenario lacks corresponding quantitative benchmarks. The scalp is a sensitive organ with heterogeneous tolerance across regions (e.g., temples, nape, crown), yet no systematic measurements exist for defining safe force, torque, or thermal exposure levels during hair cutting or styling. Similarly, acceptable latency ranges for remote operation have not been experimentally characterized, leaving uncertainty over how communication delays translate into practical risks. Future studies should therefore combine biomechanical experiments, psychophysical assessments, and user studies to establish scientifically grounded safety thresholds. These values would not only inform controller design but also provide regulators with tangible metrics for safety certification.
- **Insufficient benchmarks and validation tools.** The absence of common benchmarks makes it difficult to compare safety mechanisms across studies. Public datasets capturing hair–scalp boundaries, tool–skin interactions, and multimodal sensor signals during haircutting do not currently exist. Moreover, most experiments rely on ad-hoc test setups, limiting reproducibility. A promising direction is the development of anthropomorphic head phantoms with artificial hair and biofidelic scalp materials. Such platforms could integrate embedded sensors for ground-truth force, temperature, and vibration measurements, enabling both algorithm testing and safety validation under controlled yet realistic conditions. Similarly, simulation environments incorporating realistic hair models would support large-scale training and stress-testing of safety algorithms before physical deployment.
- **Need for personalized safety modeling.** Haircutting tasks inherently involve individual variability. Hair density, curliness, and reflectance affect visual perception, while scalp sensitivity and geometry influence safe force thresholds. A uniform safety model risks either over-restriction, reducing efficiency, or under-protection, compromising safety. Future work should therefore emphasize adaptive and personalized safety modeling. This may involve learning user-specific safety envelopes online, using multimodal sensing (vision, force, thermal), or leveraging pre-interaction calibration routines. Machine learning approaches, including meta-learning or transfer learning, could allow safety parameters to generalize across populations while adapting to individual differences in real time.
- **Towards protocols and standardization.** Although general robotic safety standards such as ISO 10218 and ISO/TS 15066 provide guidelines for industrial and collaborative robots, they do not cover the unique requirements of haircutting robots, which operate in direct proximity to the head and face and involve sharp or heated tools. Without

domain-specific protocols, system validation risks remaining inconsistent. A crucial next step is the formulation of task-oriented evaluation protocols, including standardized test cases, acceptance criteria, and reporting formats. For example, protocols could mandate tests on scalp phantoms under defined force and temperature conditions, or specify network delay thresholds for teleoperated modes. Such efforts can serve as a foundation for industry-wide standardization, gradually extending the safety framework into formal regulations.

- **Ethical and privacy considerations.** Beyond physical safety, haircutting robots inevitably involve the collection and processing of sensitive data, such as facial geometry, scalp features, and hairstyle preferences. These data raise privacy concerns related to storage, sharing, and potential misuse, particularly when biometric recognition techniques are employed. Ethical questions also emerge regarding autonomy versus human oversight, liability in the event of harm, and cultural acceptance of robotic interaction with personal appearance. Addressing these issues will require not only technical safeguards such as on-device processing and data anonymization, but also transparent policies and regulatory oversight.

Overall, bridging these gaps will require interdisciplinary efforts across robotics, biomechanics, human factors, and ethics. Addressing them will pave the way for haircutting robots to evolve from experimental prototypes into trustworthy service systems, enabling safe and acceptable deployment in real-world environments.

8 | CONCLUSION

Robotic haircutting combines the complexity of physical human–robot interaction with task-specific hazards that directly affect user safety and comfort. This paper has outlined how risks emerge across mechanical structures, sensing systems, and algorithmic control, and how lessons from related domains can inform their mitigation. Beyond technical safeguards, the path toward real deployment requires the establishment of measurable safety thresholds, reproducible testing platforms, and protocols that reflect the unique demands of haircutting tasks. Addressing these issues will enable haircutting robots to evolve from experimental prototypes into practical service systems. Achieving this transition is essential not only for advancing robotics research but also for shaping public acceptance of robots performing intimate, high-precision tasks in everyday life.

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None reported.

CONFLICT OF INTEREST
The authors declare no potential conflict of interests.

REFERENCES

1. Cherubini A, Passama R, Crosnier A, Lasnier A, Fraisse P. Collaborative manufacturing with physical human–robot interaction. *Robotics and Computer-Integrated Manufacturing.* 2016;40:1–13.
2. Mohebbi A. Human-robot interaction in rehabilitation and assistance: a review. *Current Robotics Reports.* 2020;1(3):131–144.
3. Haddadin S, Croft E. Physical human–robot interaction. In: , , Springer, 2016:1835–1874.
4. Li S. *Haircutting Robots*. Springer, 2025.
5. Khan AT, Li S. Robotic Haircutting Systems: A Survey of Methods, Challenges, and Hair Modeling Insights. *Journal of Selected Areas in Sensors.* 2025.
6. Li S. Haircutting Robots: From Theory to Practice. *Automation.* 2025;6(3).
7. De Santis A, Siciliano B, De Luca A, Bicchi A. An atlas of physical human–robot interaction. *Mechanism and Machine Theory.* 2008;43(3):253–270.
8. Alami R, Albu-Schäffer A, Bicchi A, et al. Safe and dependable physical human-robot interaction in anthropic domains: State of the art and challenges. In: IEEE. 2006:1–16.
9. Selvaggio M, Cognetti M, Nikolaidis S, Ivaldi S, Siciliano B. Autonomy in physical human-robot interaction: A brief survey. *IEEE Robotics and Automation Letters.* 2021;6(4):7989–7996.
10. Kirschner RJ, Micheler CM, Zhou Y, et al. Towards safe robot use with edged or pointed objects: A surrogate study assembling a human hand injury protection database. In: IEEE. 2024:12680–12687.
11. SMBPB S, Valori M, Legnani G, Fassi I. Assessing Safety in Physical Human–Robot Interaction in Industrial Settings: A Systematic Review of Contact Modelling and Impact Measuring Methods. *Robotics.* 2025;14(3):27.
12. Halme RJ, Lanz M, Kämäräinen J, Pieters R, Latokartano J, Hietanen A. Review of vision-based safety systems for human-robot collaboration. *Procedia Cirp.* 2018;72:111–116.
13. Secil S, Ozkan M. Minimum distance calculation using skeletal tracking for safe human-robot interaction. *Robotics and Computer-Integrated Manufacturing.* 2022;73:102253.
14. Xue T, Wang W, Ma J, Liu W, Pan Z, Han M. Progress and prospects of multimodal fusion methods in physical human–robot interaction: A review. *IEEE Sensors Journal.* 2020;20(18):10355–10370.
15. Papanastasiou S, Kousi N, Karagiannis P, et al. Towards seamless human robot collaboration: integrating multimodal interaction. *The International Journal of Advanced Manufacturing Technology.* 2019;105(9):3881–3897.
16. Fang C, Peternel L, Seth A, Sartori M, Mombaur K, Yoshida E. Human modeling in physical human-robot interaction: A brief survey. *IEEE Robotics and Automation Letters.* 2023;8(9):5799–5806.
17. Ahmed MR. *Compliance control of robot manipulator for safe physical human robot interaction*. PhD thesis. Örebro university, 2011.
18. Iroju O, Adeniyi OO, Ikono R. State of the art: a study of human-robot interaction in healthcare. 2017.
19. Polygerinos P, Correll N, Morin SA, et al. Soft robotics: Review of fluid-driven intrinsically soft devices; manufacturing, sensing, control, and applications in human-robot interaction. *Advanced engineering materials.* 2017;19(12):1700016.
20. Qaddoori Fenjan S, Fathollahi Dehkordi S. Soft robotic system with continuum manipulator and compliant gripper: design, fabrication, and implementation. In: . 13. MDPI. 2024:298.
21. Mohammadi Nejad Rashty A, Sharbafi MA, Mohseni O, Seyfarth A. Role of compliant mechanics and motor control in hopping-from human to robot. *Scientific Reports.* 2024;14(1):6820.
22. Ahamed R, Choi SB, Ferdaus MM. A state of art on magneto-rheological materials and their potential applications. *Journal of*

Intelligent material Systems and structures. 2018;29(10):2051–2095.
23. International Organization for Standardization . ISO/TS 15066:2016 Robots and robotic devices – Collaborative robots. 2016. [Online]. Available: https://www.iso.org/standard/62996.html.
24. International Organization for Standardization . ISO 13482:2014 Robots and robotic devices – Safety requirements for personal care robots. 2014. [Online]. Available: https://www.iso.org/standard/53820.html.
25. International Organization for Standardization . ISO 10218-1:2025 Robotics — Safety requirements – Part 1: Industrial robots. 2025. [Online]. Available: https://www.iso.org/standard/73933.html.
26. International Organization for Standardization . ISO 10218-2:2025 Robotics — Safety requirements – Part 2: Industrial robot applications and robot cells. 2011. [Online]. Available: https://www.iso.org/standard/73934.html.
27. International Organization for Standardization . ISO 18646-1:2016 Robotics – Performance criteria for service robots – Part 1: Locomotion for wheeled robots. 2016. [Online]. Available: https://www.iso.org/standard/63127.html.
28. International Organization for Standardization . ISO 18646-2:2024 Robotics – Performance criteria for service robots – Part 2: Navigation. 2016. [Online]. Available: https://www.iso.org/standard/82643.html.
29. International Electrotechnical Commission . IEC 60601-1:2005 Medical electrical equipment – Part 1: General requirements for basic safety and essential performance. 2005. [Online]. Available: https://webstore.iec.ch/en/publication/2606.
30. International Electrotechnical Commission . IEC 80601-2-XX: Medical electrical equipment – Particular requirements for basic safety and essential performance of home healthcare environment devices. 2017. [Online]. Available: http://www.iec.ch.
31. International Organization for Standardization . ISO 14971:2019 Medical devices – Application of risk management to medical devices. 2019. [Online]. Available: https://www.iso.org/standard/72704.html.
32. European Union . Directive 2006/42/EC of the European Parliament and of the Council on machinery, and amending Directive 95/16/EC. 2006. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:32006L0042.
33. Law T, Scheutz M. Trust: Recent concepts and evaluations in human-robot interaction. *Trust in human-robot interaction.* 2021:27–57.
34. Hancock PA, Billings DR, Schaefer KE, Chen JY, De Visser EJ, Parasuraman R. A meta-analysis of factors affecting trust in human-robot interaction. *Human factors.* 2011;53(5):517–527.
35. Yagoda RE, Gillan DJ. You want me to trust a ROBOT? The development of a human–robot interaction trust scale. *International Journal of Social Robotics.* 2012;4(3):235–248.
36. Rubagotti M, Tusseyeva I, Baltabayeva S, Summers D, Sandygulova A. Perceived safety in physical human–robot interaction—A survey. *Robotics and Autonomous Systems.* 2022;151:104047.
37. Singh S, Bora GS, Devana SS, Mavuduru RS, Singh SK, Mandal AK. Instrument malfunction during robotic surgery: A case report. *Indian Journal of Urology.* 2016;32(2):159–160.
38. Rajih E, Tholomier C, Cormier B, et al. Error reporting from the da Vinci surgical system in robotic surgery: A Canadian multispecialty experience at a single academic centre. *Canadian Urological Association Journal.* 2017;11(5):E197.
39. Kim WT, Ham WS, Jeong W, Song HJ, Rha KH, Choi YD. Failure and malfunction of da Vinci Surgical systems during various robotic surgeries: experience from six departments at a single institute. *Urology.* 2009;74(6):1234–1237.
40. Han D, Park MY, Choi J, Shin H, Behrens R, Rhim S. Evaluation of force pain thresholds to ensure collision safety in worker-robot collaborative operations. *Frontiers in Robotics and AI.* 2024;11:1374999.
41. Jaiswal S, Jaiswal A. THERMAL CONDUCTIVITY OF MATERIALS: IMPLICATIONS FOR HEAT MANAGEMENT IN MEDICAL DEVICES. *ShodhKosh: Journal of Visual and Performing Arts.* 2024;5. doi: 10.29121/shodhkosh.v5.i7.2024.2066
42. Behrens R, Pliske G, Umbreit M, Piatek S, Walcher F, Elkmann N. A statistical model to determine biomechanical limits for physically safe interactions with collaborative robots. *Frontiers in Robotics and AI.* 2022;8:667818.
43. Martin N, Falder S. A review of the evidence for threshold of burn injury. *Burns.* 2017;43(8):1624–1639.
44. Madej M, Piotrowska K, Vicen M, Zatkaliková V. Wear and Corrosion Behavior of Diamond-like Carbon Coatings in Artificial Saliva. *Coatings.* 2025;15(3):305.
45. The Economic Times . An industrial robot fails to differentiate between a human and a box, kills a man. 2023. Accessed: 2025-09-14.
46. BBC News . Boeing 737 Max: What went wrong?. 2019. Accessed: 2025-09-14.
47. The Verge . Waymo self-driving car incident in Arizona raises questions about sensor coverage. 2018. Accessed: 2025-09-14.
48. The New York Times . Self-driving Uber car kills pedestrian in Arizona, where robots roam. 2018. Accessed: 2025-09-14.
49. U.S. FDA . MAUDE – Manufacturer and User Facility Device Experience: da Vinci system incident reports. n.d. Accessed: 2025-09-14.
50. Hainsworth S, Delaney R, Rutty GN. How sharp is sharp? Towards quantification of the sharpness and penetration ability of kitchen knives used in stabbings. *International journal of legal medicine.* 2008;122(4):281–291.
51. Liu Z, Chen H, Sui J, et al. Failure behavior and influence of surgical tool edges in soft tissue cutting. *Journal of Manufacturing Processes.* 2021;68:69–78.
52. Krenik MW. Automated hair cutting system and method of operation thereof. 2018. US Patent 9,925,676.
53. Feizi N, Atashzar SF, Kermani MR, Patel RV. Design and modeling of a smart torque-adjustable rotary electroadhesive clutch for application in human–robot interaction. *IEEE/ASME Transactions on Mechatronics.* 2023;28(5):2738–2748.
54. Zheng J, Wang S, Housden J, Hou ZG, Singh D, Rhode K. A safety joint with passive compliant and manual override mechanisms for medical robotics. In: IEEE. 2021:1–4.
55. Robinson DW, Pratt JE, Paluska DJ, Pratt GA. Series elastic actuator development for a biomimetic walking robot. In: IEEE. 1999:561–568.
56. McCarthy CT, Hussey M, Gilchrist MD. An investigation into scalpel blade sharpness using cutting experiments and finite element analysis. *Key Engineering Materials.* 2005;293:769–776.
57. Dreher DW. Hair Dryer Assembly. 1976. U.S. Patent.
58. ST-12 Bimetal Thermal Cutout Switch for Hair Dryer. https://thermostat-china.en.made-in-china.com/product/KsHEAXorXYhc/China-Thermal-Cutout-Switch-for-Hair-Dryer-St-12-Thermostat.html; . Accessed: 2025-09-18.
59. Zhao B, Nelson CA. Sensorless force sensing for minimally invasive surgery. *Journal of medical devices.* 2015;9(4):041012.
60. Lu Z, Gao X, Yu H. GTac: A biomimetic tactile sensor with skin-like heterogeneous force feedback for robots. *IEEE Sensors Journal.* 2022;22(14):14491–14500.
61. Kappassov Z, Baimukashev D, Kuanyshuly Z, Massalin Y, Urazbayev A, Varol HA. Color-coded fiber-optic tactile sensor for an elastomeric robot skin. In: IEEE. 2019:2146–2152.
62. Ionut-Cristian S, Dan-Marius D. Using inertial sensors to determine head motion—A review. *Journal of Imaging.* 2021;7(12):265.
63. Borowska-Terka A, Strumiłło P. A yaw tracking algorithm for head movement from inertial sensors data. *Metrology and Measurement Systems.* 2023:737–754.
64. Mur-Artal R, Tardós JD. Orb-slam2: An open-source slam system for monocular, stereo, and rgb-d cameras. *IEEE transactions on robotics.* 2017;33(5):1255–1262.

65. Tsuji S, Kohama T. Proximity skin sensor using time-of-flight sensor for human collaborative robot. *IEEE Sensors Journal.* 2019;19(14):5859–5864.
66. Valkenburg RJ, McIvor AM. Accurate 3D measurement using a structured light system. *Image and Vision Computing.* 1998;16(2):99–110.
67. Fonseca D, Safeea M, Neto P. A flexible piezoresistive/self-capacitive hybrid force and proximity sensor to interface collaborative robots. *IEEE Transactions on Industrial Informatics.* 2022;19(3):2485–2495.
68. Wade J, Bhattacharjee T, Williams RD, Kemp CC. A force and thermal sensing skin for robots in human environments. *Robotics and Autonomous Systems.* 2017;96:1–14.
69. Afzal U, Afzal F, Maryam K, Aslam M. Fabrication of flexible temperature sensors to explore indeterministic data analysis for robots as an application of Internet of Things. *RSC advances.* 2022;12(27):17138–17145.
70. Coşar S, Yan Z, Zhao F, Lambrou T, Yue S, Bellotto N. Thermal camera based physiological monitoring with an assistive robot. In: IEEE. 2018:5010–5013.
71. Li H, Tian S, Yang Z. A novel blade vibration monitoring experimental system based on blade tip sensing. *Materials.* 2022;15(19):6987.
72. Zhang P, Gao D, Lu Y, Ma Z, Wang X, Song X. Cutting tool wear monitoring based on a smart toolholder with embedded force and vibration sensors and an improved residual network. *Measurement.* 2022;199:111520.
73. Wang Y, Zhao J, Xia Y, Liu P. Configuration optimization of bionic piezoelectric hair sensor for acoustic/tactile detection. *Bioinspiration & biomimetics.* 2020;15(5):056015.
74. NTSB . Collision Between Vehicle Controlled by Developmental Automated Driving System and Pedestrian. Tech. Rep. NTSB/HAR-19/03, National Transportation Safety Board; 2019. Accessed: 2025-09-14.
75. The Verge . Tesla is under investigation for delayed reporting of Autopilot/FSD crashes. 2025. Accessed: 2025-09-14.
76. U.S. FDA . Class 2 Device Recall: da Vinci Xi Surgical System (P5 software anomaly). 2017. Accessed: 2025-09-14.
77. University of Washington News . UW researchers hack a teleoperated surgical robot to reveal security flaws. 2015. Accessed: 2025-09-14.
78. Quist E, others . Telesurgical Robotic Operative Network (TRON) – Final Report. tech. rep., TATRC; 2023. Accessed: 2025-09-14.
79. Reuters . U.S. crash involving Tesla Autopilot under investigation. 2016. Accessed: 2025-09-14.
80. Masalskyi V, Dzedzickis A, Korobiichuk I, Bučinskas V. Hybrid mode sensor fusion for accurate robot positioning. *Sensors.* 2025;25(10):3008.
81. Saleem Z, Gustafsson F, Furey E, McAfee M, Huq S. A review of external sensors for human detection in a human robot collaborative environment. *Journal of Intelligent Manufacturing.* 2025;36(4):2255–2279.
82. Khattak S, Papachristos C, Alexis K. Vision-depth landmarks and inertial fusion for navigation in degraded visual environments. In: Springer. 2018:529–540.
83. Han X, Chen S, Fu Z, et al. Multimodal fusion and vision-language models: A survey for robot vision. *arXiv preprint arXiv:2504.02477.* 2025.
84. Sheridan TB. Space teleoperation through time delay: Review and prognosis. *IEEE Transactions on robotics and Automation.* 2002;9(5):592–606.
85. Periotto R, Ferizbegovic M, Barbosa FS, Sundin RC. MPC-CBF with adaptive safety margins for safety-critical teleoperation over imperfect network connections. In: IEEE. 2024:1609–1615.
86. Uyulan C. Robust passivity-based nonlinear controller design for bilateral teleoperation system under variable time delay and variable load disturbance. *Nonlinear Engineering.* 2024;13(1):20220358.
87. Risiglione M, Sleiman JP, Minniti MV, Cizmeci B, Dresscher D, Hutter M. Passivity-based control for haptic teleoperation of a legged manipulator in presence of time-delays. In: IEEE. 2021:5276–5281.
88. Periotto R. Safety-Critical Teleoperation with Time-Varying Delays: MPC-CBF-based approaches for obstacle avoidance. 2023.
89. Yaacoub JPA, Noura HN, Salman O, Chehab A. Robotics cyber security: Vulnerabilities, attacks, countermeasures, and recommendations. *International Journal of Information Security.* 2022;21(1):115–158.
90. Iqbal S, Farooq S, Shahzad K, Malik AW, Hamayun MM, Hasan O. SecureSurgiNET: A framework for ensuring security in telesurgery. *International Journal of Distributed Sensor Networks.* 2019;15(9):1550147719873811.
91. Alemzadeh H, Chen D, Li X, Kesavadas T, Kalbarczyk ZT, Iyer RK. Targeted attacks on teleoperated surgical robots: Dynamic model-based detection and mitigation. In: IEEE. 2016:395–406.
92. Sobhi S, Reshadi M, Zarft N, Terheide A, Dick S. Condition monitoring and fault detection in small induction motors using machine learning algorithms. *Information.* 2023;14(6):329.
93. Eang C, Lee S. Predictive maintenance and fault detection for motor drive control systems in industrial robots using CNN-RNN-based observers. *Sensors.* 2024;25(1):25.
94. Zhang Y, Wu J, Gao B, et al. Fault Types and Diagnostic Methods of Manipulator Robots: A Review. *Sensors.* 2025;25(6):1716.
95. Raouf I, Lee H, Kim HS. Mechanical fault detection based on machine learning for robotic RV reducer using electrical current signature analysis: A data-driven approach. *Journal of Computational Design and Engineering.* 2022;9(2):417–433.
96. Khatib O. Real-time obstacle avoidance for manipulators and mobile robots. *The international journal of robotics research.* 1986;5(1):90–98.
97. Borenstein J, Koren Y. Real-time obstacle avoidance for fast mobile robots. *IEEE Transactions on systems, Man, and Cybernetics.* 1989;19(5):1179–1187.
98. Pandey A, Pandey S, Parhi D. Mobile robot navigation and obstacle avoidance techniques: A review. *Int Rob Auto J.* 2017;2(3):00022.
99. Latombe JC. *Robot motion planning*. 124. Springer Science & Business Media, 2012.
100. Elbanhawi M, Simic M. Sampling-based robot motion planning: A review. *Ieee access.* 2014;2:56–77.
101. Canny J. *The complexity of robot motion planning*. MIT press, 1988.
102. Vasic M, Billard A. Safety issues in human-robot interactions. In: IEEE. 2013:197–204.
103. Xiao W, He T, Dolan J, Shi G. Safe deep policy adaptation. In: IEEE. 2024:17286–17292.
104. Li F, Sun Y, Zhao W, Chen R, Wei T, Liu C. Learn with imagination: Safe set guided state-wise constrained policy optimization. *arXiv preprint arXiv:2308.13140.* 2023.
105. Yang Q, Simão TD, Tindemans SH, Spaan MT. Safety-constrained reinforcement learning with a distributional safety critic. *Machine Learning.* 2023;112(3):859–887.
106. Wachi A, Sui Y. Safe reinforcement learning in constrained markov decision processes. In: PMLR. 2020:9797–9806.